De La Salle University
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System Administration Laboratory Manual

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Section:	S18	Grade:	

3.0 Unix / Linux Command Line Interface (CLI) Basics –File Utilities, Redirection and Pipes

The Unix system provides file utilities (or commands) that one can use. File utilities can used to display file content, check file type, count words or lines in a file, sort words or look for key words using regular expression. Aside from file utilities, the Unix system also provides facilities for redirection and pipes. Redirection allows output of a command to be "redirected" to a file. Pipes allows the output of the command as an input to another command. Unix or Linux also has a command to display information or help of a command using the "`man`" command. The objectives of this section are:

- Be able to use software utilities to manipulate or display file content
- Be able to use redirection to save output of a command to file or display output to a terminal
- Be able to use pipes to "combine" commands by using output of a command as input to another command

3.1. Utilities

3.1.1. On-line Help

The Unix environment provides a simple on-line help system. The command "`man`" is used to browse the help pages (manuals). This help system is sometimes referred to as "man pages". Typing "`man ls`" at the command prompt will show you an output in Fig. 3.1:

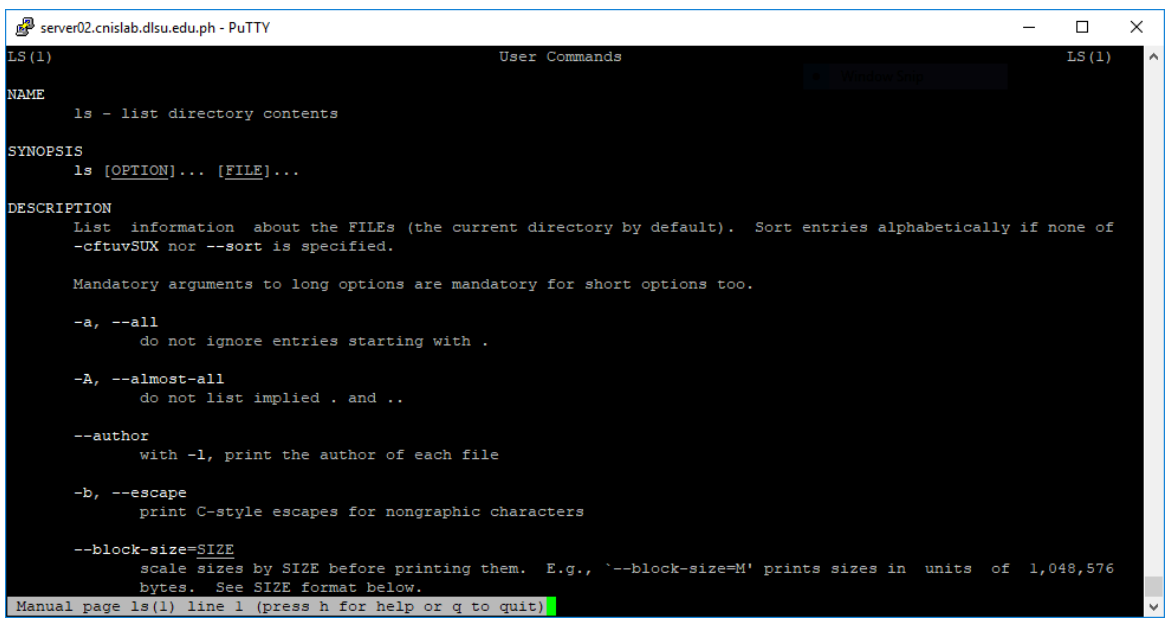

Fig. 3.1 man page for the "`ls`" command

The man page most of the time will contain the following sections (different Unix versions or authors of man pages sometimes add sections): name, synopsis, description, and see also. The name section shows the name of the command and alternative commands that does the same thing. The synopsis section shows the syntax of the command while the description gives a short text on what the command does and the options that can be used. The see also section gives other related commands.

Man pages also provide help on programming libraries, utilities, file formats and other topics on the Unix environment.

Step 1 On the command prompt, type the command "`man ls`".

1a. What does the given command do?

The command provides information on the 'ls' command. This is because of 'man', which stands for manual. With this command, we can learn more about the function of a certain command.

1b. Press the spacebar key. What does it do?

The spacebar scrolls the page down. We get to see the lines towards the end.

1c. Press the "`f`" key. What does it do? Is it the same as the space bar key?

It also scrolls the page down. It is the same as the space bar key

1d. Press the "`b`" key. What does it do?

The command scrolls the page up, it allows us to see the lines at the beginning.

1e. Type the following "`/format`" and press enter. What does it do?

It highlights all instances of 'format' on the man page. We can use this to highlight important information we want to see.

1f. Press the "`q`" key. What does it do?

It exits the command and brings us back to the main page

Step 2 On the command prompt, type the command "`man ls`". Scroll the pages and check the sections of the man page.

2a. Can you identify the sections mentioned? List down the main sections you see.

The sections I identified are Name, Synopsis, Description, Author, Reporting Bugs, Copyright, and See also.

2b. Write down the description of the "`ls`" command (do not include the options).

List information about the FILEs (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.

2c. Exit the man page.

Step 3 On the command prompt, type the command "`man -f ls`".

3a. Describe the output of the given command. What is the `-f` option for?

The output is a one-line statement which shows that there is one ls manual page. -f is used to show man pages that match the string

Step 4 On the command prompt, type the command "`man -k ls`".

4a. Describe the output of the given command.

The output seems to be a long list of commands and the functions they perform. A commonality is that all of the commands and descriptions contain 'ls' in them.

4b. What is the "`-k`" option for?

The -k option is used to search for man pages that have words matching the given command.

If you look carefully at the upper right corner of the man page, there is a "( )" after the command. This indicates in what section of the command are you viewing. Using the "s" option allows you to specify which section would you want to view. The `man` utility displays the lowest numbered section by default when no section number is provided from the command.

3.1.2. The "file" and "wc" Commands

The "`file`" command allows the user to determine the file type. This is useful if you cannot remember what the file is for. The "`wc`" command prints the number of lines, words, and bytes in a file.

Step 5 At the command prompt, inside your home directory, enter the following commands:

"`cp /home/files/rfc2186.txt rfc2186.txt`" and

"`cp /home/files/rfc2186.pdf rfc2186.pdf`". Verify if the files were copied.

5a. At the command prompt, enter the command "`file rfc2186.txt`". What type of file is it?

It is an ASCII text

5b. At the command prompt, enter the command "`file rfc2186.pdf`". What type of file is it?

It is a PDF document, version 1.4, 9 pages

5c. At the command prompt, enter the command "`mv rfc2186.txt rfc2186`". After renaming the file, enter the command "`file rfc2186`". Was it still able to determine the file type?

Yes, the command's result indicates it is an ASCII text

Step 6 At the command prompt, enter the command "`wc rfc2186`".

6a. Describe the output of the command.

The output is the following numbers followed by its filename, "507 2313 18808 rfc2186". These numbers indicate the number of lines, words, and bytes the file has.

6b. How many lines, words, and bytes does the file have?

Lines – 507
Words – 2313
Bytes – 18808

6c. At the command prompt, enter the command "`wc -c rfc2186`". How many bytes does the file have? Is it the same as its file size?

It has 18808 bytes. It is still the same file size, the -c option simply shows the number of bytes the file contains only.

6d. At command prompt, enter the command "`wc rfc2186.pdf`". How many lines, words, and bytes does the file have?

Lines – 672
Words – 1933
Bytes – 21507

3.1.3. The "`more`" and "`cat`" Commands

The "`cat`" command can concatenate files and print on standard output (which is usually the screen). This is useful for checking the contents of short file, especially script files (will be tackled later). The command can accept more than one filename as an argument that can be printed on the standard output.

The problem with the "`cat`" command is that if the file is too long, the displaying of the file will just flash by. An alternative command is the "`more`" command. This command displays the file on the standard output and provides scrolling capability. Commands for scrolling are the same in the "`man`" utility.

Step 7 At the command prompt, enter the command "`cp /home/files/fruits fruits`" and press enter. After copying the file, enter the command "`cp /home/files/vegetables vegetables`" and press enter. Verify if you have copied the file using the "`ls`" command.

7a. After verifying the files were copied, enter the command "`cat fruits`". What did the command do?

The command's result shows the content of the 'fruits' file. It is a list of fruits.

7b. Enter the command "`cat vegetables`". What did the command do?

This time it showed the content of the 'vegetables' file. It is a list of vegetables.

7c. Enter the command "`cat fruits vegetables`". What did the command do?

It showed the content of both files. Our output is a list of both fruits and vegetables. A key thing I noticed is that the fruits were shown before the vegetables

7d. Enter the command "`cat vegetables fruits`". What did the command do? Is it different from the previous command? What can you generalize about this command and the command from the previous question?

It also showed the content of both files. However, this time the list shows the vegetables first before the fruits. We can generalize that the order of content that is shown depends on the order of the files in the command

7e. Enter the command "`cat rfc2186`". What did the command do? Were any blocks of text left on the screen?

The command showed the content of rfc2186. However, the file is very long to the point no blocks of text was left on the screen. To view the content, I need to scroll all the way up.

Step 8 At the command prompt, enter the command "`more rfc2186`" and press enter.

8a. Are you able to view the file?

Yes we are able to view the file. This time the screen is not blank and shows the start of the file content.

8b. Press the spacebar key. What does it do?

It scrolls the screen up. It brings us towards the end of the file.

8c. Press the "`f`" key. What does it do? Is it the same as the space bar key?

Yes f and the space bar key are similar in function. They both scroll the screen up.

8d. Press the "`b`" key. What does it do?

It scrolls the screen down. It brings us towards the start of the file.

8e. Press the "`q`" key. What does it do?

It exits the file and brings back the prompt. This key is used when we are done reading the file and want to type a new command.

There are still other commands to scroll the screen in the "`more`" command. Use the man pages for the "`more`" command to check the scroll commands and other options.

If your screen looks confusing, type in the command "`clear`" to blank out the display.

3.1.4. The "`head`" and "`tail`" Commands

The "head" command outputs the first few lines of a few on the screen while the "tail" command outputs the last few lines of a file on the screen. The default number of line output is ten (10). These commands useful for checking log files, especially to network administrators.

Step 9 At the command prompt, type the command "`cp /home/files/PH_Cities.txt PH_Cities.txt`" then type in the command "`head PH_Cities.txt`".

9a. What was the output of the given command? How is it different from the output of "`cat PH_Cities.txt`"?

The given command listed the first 10 lines of the file's content. 'head' and 'cat' differ as 'head' lists the first 10 lines of a file's content, while 'cat' lists the whole file content

9b. At the command prompt, type the command "`head -c 20 PH_Cities.txt`". What was the output?

The output printed 2 lines only. It showed Angeles and Antipolo.

9c. At the command prompt, type the command "`head -n 20 PH_Cities.txt`". What was the output? How was it different from the output in the previous question?

The output is 20 lines of the file's content. It listed the cities starting from Angeles to Pasay. It differs greatly as more cities were shown using -n 20 rather than -c 20.

9d. At the command prompt, type the command "`tail PH_Cities.txt`". What was the output?

It shows the last 10 lines of the file's content. It showed cities Gapan to Imus

9e. At the command prompt, type the command "`tail -c 20 PH_Cities.txt`". What was the output?

aytay
Alaminos
Imus

This is the output. It started from the last line and worked upwards. Tagaytay was cut off because this is what 20 bytes of the file would contain.

9f. At the command prompt, type the command "`tail -n 20 PH_Cities.txt`". What was the output? How was it different from the output in the previous question?

The output is the last 20 lines of the file's content. It showed cities from Las Pinas to Imus. They differ as they output different number of lines.

9g. What is the difference between options "`c`" and "`n`"?

With head and tail, the -c [num of bytes] option will output the file's content only up to the number of bytes stated. Whereas, -n [num of lines] option will output the file's content up to the number of lines stated. They would mostly have different outputs as -c treats the number as bytes, while -c treats the number as the number of lines.

3.1.5. The "grep" Command

The "`grep`" command searches and prints the lines in a file with a matching pattern. This is useful for searching patterns in a text or binary file. Most of the time, network administrators use this to search if a certain process is running (to be discussed later) or searching for keywords in a configuration file (again, to be discussed later).

Step 10 At the command prompt, enter the command "`grep apple fruits`".

10a. What was the output? Why do you think the output is like that?

The output showed 'apple' in red font. I believe the output is like that as our pattern matches 'apple' in the fruits file, it is red in color to highlight a match.

10b. At the command prompt, enter the command "`grep broccoli fruits`". Was there an output? Why?

There is not output because there is no broccoli in the fruits file.

10c. At the command prompt, enter the command "`grep broccoli vegetables`". Was there an output? Why?

No there is no output. Upon checking the vegetables file contains 'brocolli' not 'broccoli'. When I did the command 'grep broccoli vegetables', the output is 'brocolli' in red font.

3.1.6. The "sort" Command

The "`sort`" command allows a user to sort alphanumeric text in a file. If the text file has columns, the "`sort`" command is able to sort by column. This is useful if you have a spreadsheet like a text file.

Step 11 At the prompt, enter the command "`cat fruits`". Observe the output of the command.

11a. Are the contents listed in alphabetical order?

No they are not in alphabetical order.

11b. At the prompt, enter the command "`sort fruits`". Observe the output of the command. Compare the output of this command to the output in the previous question.

Now the output is sorted in alphabetical order

Step 12 At the prompt, enter the command "`cp /home/files/fruits2 fruits2`".

12a. After copying the file, enter the command "`cat fruits2`". Observe the output of the command. What was the output?

The output is as follows:
mango
banana
papaya
tomato
vanilla

I observed that they are not in alphabetical order.

12b. At the prompt, enter the command "`sort fruits2`". Observe the output of the command. What was the output?

The output is now sorted in alphabetical order

12c. At the prompt, enter the command "`sort -r fruits2`". Observe the output of the command. What was the output? How is it different from the previous question?

The output are sorted alphabetically in reverse from z to a. It differs because of the -r option which stands for reverse.

12d. At the prompt, enter the command "`sort -k 2 fruits2`". Observe the output of the command. What was the output? How is it different from the command in step 12b?

The output is similar to the output in step 12b. The -k option indicates sorting by field number. Since the number is 2, it will sort using the second word (next word after the first space). However since our list only contain 1 word per line, this option has no effect.

12e. At the prompt, enter the command "`sort -r -k 2 fruits2`". Observe the output of the command. What was the output? How is it different from the command in the previous step?

The output is similar to the output in step 12c. It is arranged alphabetically in reverse.

3.2. Redirection and Pipes

Redirectors "redirect" the standard output of a command to a file or device file. This is useful for saving a long list of directories and files in text file for example. A redirection command looks like:

`command > file` or `command > file_device`

The ">" is the redirection symbol. Redirection symbols are:

Redirection Symbol	Description
>	Sends output of a command to another file or file device
<	Sends file or file device to a command
>>	Appends an output of a command to another file or file device.

Pipes are the same as redirection but use the output of a command as input to another command. The pipe symbol is "|".

Step 13 At the command prompt, enter the command "`ls -l > list`".

13a. Was the text file `list` created?

Yes the text file was created.

13b. Use the "`cat`" or "`more`" command to view the "`list`" text file. Describe the file contents.

The 'list' file contains a list of the file contents from the home directory. Since -l was used, the content used the long listing format which includes other information other than the file name.

13c. At the command prompt, enter the command "`ls -l > list`". Was a new text file list created? What does the text file contain? Was it the same as before?

Upon checking using the command 'cat list', the content still looks the same. However, the time at which the file 'list' was created changed. It is now the latest time created. This means that a new 'list' file was created and the previous 'list' was overridden.

13d. At the command prompt, enter the command "`ls -l >> list`". Was a new text file list created? What does the text file contain? Was it the same as before?

It is not the same as before. The output shows two blocks of the same files. It seems that the output of the command was appended to the previous content of the file.

13e. What is the difference between the ">" and the ">>" redirection symbol?

They both send the output of a command to a file. However, ">" will override the previous content and ">>" will append the previous content.

Step 14 At the command prompt, enter the command "`cd /etc/`". After changing to the "`/etc`" directory, enter the command "`ls -l`".

14a. In terms of the number of lines displayed, are all files displayed immediately when you used "`ls -l`"?

Yes all the files were displayed immediately

14b. At the command prompt, enter the command "`ls -l | more`". Were you able to see all files immediately?

No, I needed to use the space bar key to expand the list to view the rest of the files.

14c. How many commands were used in the previous question (14b)? What is/are the command/s used?

We have two commands. They are 'ls', which is used to list a file's content and 'more' which shows a file's content part-by-part. -l is an option.

Remember that the "`more`" in the command is neither an option nor an argument. "`more`" is a command utility that is used to view a file.

Step 15 Return to your home directory (using "`cd ~`") then enter the command "`mkdir dir1 dir2`".

15a. At the prompt, enter the command "`ls -l`". Observe the output. What was the output of the command?

The output shows the list of files and directories in the home directory in long list format. It includes the two new directories we created.

15b. At the prompt, enter the command "`ls -l | sort -k 1`". Observe the output. How is it different from the output of the previous question (15a)?

The output is sorted using the 2nd column, which refers to the number of hard links to the file/directory. They differ as the output lists all the files first, and the directories last.

15c. At the prompt, enter the command "`ls -l | sort -r -k 1`". Observe the output. How is it different from the output of the previous question (15b)?

This time the directories were listed before the files. It is similar to 15b but it is now in reverse.

15d. At the prompt, enter the command "`ls -l | sort -k 8`". Observe the output. How is it different from the output of question 15b?

This time, the files and directories are sorted by the time created starting from earliest to latest. The date is not considered because -k 8 refers to sorting using the 8th column only, which is the time.

15e. At the prompt, enter the command "`ls -l |sort -r -k 8`". Observe the output. How is it different from the output of the previous question (15d)?

It is similar to the previous command but it sorts the files and directories in reverse. Starting from the latest to the earliest.

Remember that the "`sort`" in the command is neither an option nor an argument. "`sort`" is a command utility that is used to sort text in a file.

3.3. Conclusion/Reflection

Challenge Question 1:

View the contents of the `/home/files/file_list` file. You are asked to display the files inside while sorted from smallest file size to largest. You used the command "`cat /home/files/file_list | sort -k 5`", however, it did not produce the output you were expecting.

Output of "`cat /home/files/file_list | sort -k 5`":

```
ksolomon@sys_svr1:~$ cat /home/files/file_list | sort -k 5
total 212
-rw-r--r-- 1 root root 131676 Mar 23 08:16 rfc1034.pdf
-rw-r--r-- 1 root root    151 Apr 21 00:07 fruits2
-rw-r--r-- 1 root root  16401 Apr 20 23:52 syslog
-rw-r--r-- 1 root root  18808 Mar 23 08:16 rfc2186.txt
-rw-r--r-- 1 root root  21507 Mar 23 08:16 rfc2186.pdf
-rw-r--r-- 1 root root    254 Mar 23 08:16 PH_Cities.txt
-rw-r--r-- 1 root root     51 Mar 23 08:16 fruits
-rw-r--r-- 1 root root     56 Apr 20 23:47 vegetables
```

Expected output:

```
-rw-r--r-- 1 root root     51 Mar 23 08:16 fruits
-rw-r--r-- 1 root root     56 Apr 20 23:47 vegetables
-rw-r--r-- 1 root root    151 Apr 21 00:07 fruits2
-rw-r--r-- 1 root root    254 Mar 23 08:16 PH_Cities.txt
-rw-r--r-- 1 root root  16401 Apr 20 23:52 syslog
-rw-r--r-- 1 root root  18808 Mar 23 08:16 rfc2186.txt
-rw-r--r-- 1 root root  21507 Mar 23 08:16 rfc2186.pdf
-rw-r--r-- 1 root root 131676 Mar 23 08:16 rfc1034.pdf
```

Modify "`cat /home/files/file_list | sort -k 5`" so that the expected output is achieved. Do not create new files or use other commands aside from the given ones.

Answer:

cat /home/files/file_list | sort -k 5 -n is the answer. By adding -n, we tell the system to treat the 5th column as a number and sort them numerically from smallest to largest.

Hint: Reading the documentation will help you find the answer.

Note: `sort` is needed to sort the contents of a file. If you want to sort the output of the `ls` command directly, you can use the command "`ls -l -S -r`" for smallest to largest file size order. This is not the answer to this question.

Challenge Question 2:

Observe the output of the "`ls -l /home/files`" command:

```
ksolomon@sys_svr1:~$ ls -l /home/files
total 212
-rw-r--r-- 1 root root     51 Mar 23 08:16 fruits
-rw-r--r-- 1 root root    151 Apr 21 00:07 fruits2
-rw-r--r-- 1 root root    254 Mar 23 08:16 PH_Cities.txt
-rw-r--r-- 1 root root 131676 Mar 23 08:16 rfc1034.pdf
-rw-r--r-- 1 root root  21507 Mar 23 08:16 rfc2186.pdf
-rw-r--r-- 1 root root  18808 Mar 23 08:16 rfc2186.txt
-rw-r--r-- 1 root root  16401 Apr 20 23:52 syslog
-rw-r--r-- 1 root root     56 Apr 20 23:47 vegetables
```

Using multiple commands in succession with the pipe "|" symbol, how do you display the first five (5) files only sorted by month?

Expected output:

```
total 216
-rw-r--r-- 1 root root    151 Apr 21 00:07 fruits2
-rw-r--r-- 1 root root    437 Apr 21 00:43 file_list
-rw-r--r-- 1 root root     51 Mar 23 08:16 fruits
-rw-r--r-- 1 root root    254 Mar 23 08:16 PH_Cities.txt
-rw-r--r-- 1 root root 131676 Mar 23 08:16 rfc1034.pdf
```

Answer:

ls -l /home/files | sort -k 6 -M | head -n 6

We sort by the 6th column as that is the month and use -M to sort by Month. For the first five files, we use head -n 6 as head -n 5 will only show the first 4 files due to the line 'total 216'.